

M-MOVE-IT: Multimodal Machine Observation and Video-Enhanced Integration Tool for Data Annotation

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Abstract

M-MOVE-IT is an open-source framework that simplifies data acquisition, annotation, and AI training for wearable technology. It addresses the challenges of synchronizing video and IMU data, making it easier to develop AI models for healthcare, sports, wildlife monitoring, anti-poaching, and livestock management applications. The framework automates and streamlines managing sensors, subjects, and deployments, synchronizing data, and annotating activities. M-MOVE-IT uses the real-time clocks of sensors and an offset annotation step to achieve precise synchronization, automatically parses sensor metadata, and generates annotation tasks. The export module provides data in JSON format for easy use in AI training. M-MOVE-IT's design supports active learning and human-in-the-loop development, enhancing the efficiency and scalability of wearable technology research.

CCS Concepts

• **Software and its engineering** → **Designing software**; • **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**.

Keywords

multimodal annotation, sensor fusion, activity recognition, active learning, multimodal AI

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1 Introduction

This paper presents M-MOVE-IT – an open-source framework that simplifies data acquisition, annotation, and AI training for wearables, offering practical solutions to real-world challenges. M-MOVE-IT makes video synchronization with inertial measurement units (IMUs) data easy and can be extended to other time series data like audio or radar. This framework simplifies AI development for

wearables, revolutionizing healthcare, sports, wildlife monitoring, anti-poaching [1], and livestock management [2]. We tackle data annotation and synchronization challenges for affordable, commercially available devices, significantly enhancing the efficiency and scalability of open wearable device development. This open-source tool is designed to save valuable time for researchers and developers in open wearable technology.

Human and animal activity data provide crucial insights into behavior, well-being, and environmental interactions. Advances in small, lightweight, low-power electronics have enabled unobtrusive sensors to measure location, biometrics, and activity in humans and animals [3]. Challenges remain despite improvements in computation, memory, and battery technology allowing local sensor data processing. High-quality, annotated, and synchronized data are scarce, posing hurdles to enhancing wearable computing technologies.

2 Background and Motivation

Collecting labeled data for activity recognition is tedious and labor-intensive, typically requiring simultaneous video recording and wearable sensor data collection. The video serves as the 'ground truth,' annotated to reflect the activities of individuals wearing the sensors [4]. These annotations are then manually mapped to the sensor data to train AI models. Existing solutions focus on annotating a single modality, lack sensor synchronization support, and are not tailored for wearable applications like activity recognition. Our framework leverages domain knowledge in AI development for wearables, assisting users comprehensively. M-MOVE-IT simplifies and partially automates the process, offering a backend that supports active learning [5] and human-in-the-loop AI development. Developing AI models for wearables and multimodal AI requires integrating diverse data types such as images, videos, and sensor data. Multimodal AI enhances activity recognition by fusing video and IMU data. While too large to run on resource-constrained devices, multimodal AI models can quickly structure large datasets and improve the quality and quantity of training data by focusing human annotators on underrepresented, complex, or novel classes. However, multimodal transformer models like OpenAI's ChatGPT [6] and Meta's ImageBind [7] rely on vast quantities of well-annotated and synchronized data to interpret complex scenarios accurately. Our framework improves data annotation and synchronization, enabling advanced multimodal AI models to better understand and engage with the world.



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3 Design and Implementation

This section provides a concise overview of M-MOVE-IT, implemented with Django [8] and Python [9]. As a fork of Label Studio, M-MOVE-IT integrates collaborative annotation, campaign management, and AI-assisted labeling. Our enhancements include custom Django apps for wearable data acquisition and multimodal annotation. The M-MOVE-IT dashboard guides users through the process described below, from data acquisition to structured data ready for analysis or AI training, with steps 1-5 and 7 being entirely new and steps 6, 8, and 9 enhancing existing Label Studio features.

- (1) **Manage sensors:** Add and manage sensors for your project.
- (2) **Manage subjects:** Add and manage subjects used in your project.
- (3) **Manage deployments:** Manage sensor deployments to subjects.
- (4) **Manage data:** Add IMU and camera data to your project.
- (5) **Synchronize sensors:** Annotate and manage offsets between sensors.
- (6) **Annotate subject presence:** Annotate subject presence in your videos.
- (7) **Generate activity annotation tasks:** Automatically fuse data and create tasks to annotate data using activity labels.
- (8) **Annotate activities:** Simultaneously annotate IMU and video data using activity labels.
- (9) **Export annotations:** Export subject and activity annotations alongside synchronized multimodal sensor data.

Figure 1 shows a high-level flow diagram of M-MOVE-IT. Initially, subjects and offsets are annotated. Subject annotation is a video timeline annotation task and can be completed relatively quickly. Furthermore, this process can be automated using vision detection and tracking models.

Offsets must be annotated once per sensor pair; in our example, these are IMU-camera pairs. The offset is used throughout the project to synchronize the data. This approach offers several benefits:

- (1) No expensive synchronization equipment is required.
- (2) Syncing is done once per sensor pair and can be used for multiple recording sessions.
- (3) Sensors are synchronized when two sensor pairs share a common counter sensor.
- (4) Multiple subjects can be visually recorded using one or various cameras and move in and out of view between cameras.
- (5) Parsing information for each device type is shared in a public repository, enabling effortless parsing for known sensors.

Video segments with subject presence are automatically fused with IMU data where they overlap. M-MOVE-IT automatically generates annotation tasks with length λ that comprise the synchronized video and IMU data. A custom multimodal annotation view displays the synchronized time-series data alongside the video to aid the activity annotation effort. Due to the synchronization, any added activity label now applies to both the video and IMU data. Finally, Label Studio's machine learning (ML) back end can be used to train models for pre-labeling data and utilize active learning strategies to prioritize annotation tasks.

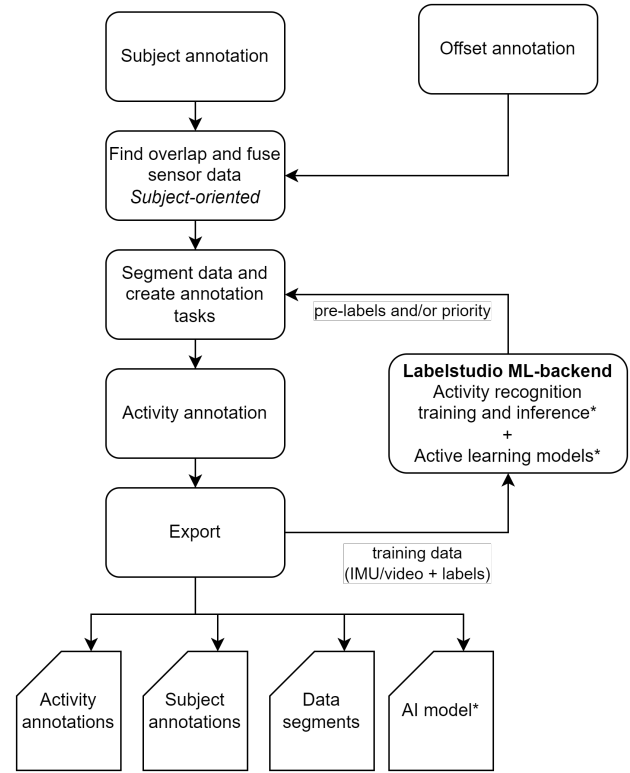


Figure 1: M-MOVE-IT high-level system flow. *Model training and export are enabled by LabelStudio's ML-backend [10]

3.1 Managing Sensors, Subjects, and Deployments

One or more wearable sensors are attached to various subjects and/or locations on those subjects to collect data for scientific projects or development purposes. Individual subject traits may be tracked to study and improve generalization. Each subject is defined with database fields such as name, size, or age to capture relevant information.

Most sensors store metadata with the sensor data, either as EXIF info in a media file or as comments in the header of a data file, often in CSV or JSON format. This metadata includes important details like datetime, location, a unique identifier, and configuration settings. Parsing information for each sensor type is required to automate the import of large amounts of files into a project.

M-MOVE-IT uses a repository containing YAML files with parsing information for each sensor type. When the user adds each sensor to the project, the available types are loaded from the repository and can be selected via a dropdown menu, as shown in Figure 2.

The subject and sensor objects are linked using deployment objects, as shown in Figure 3. The deployment interface allows the user to name the deployment, set start and end dates, and select a sensor and subject using dropdown menus. Additional information, such as the sensor's location on the subject, can also be captured.

• This field is required.

Name:

Parsable sensor id:

Sensortype:

Manual offset: seconds (integer)

Timezone: • This field is required.

Figure 2: UI to add sensor to the project

Deployments

Sensor	Subject	Begin Date Time	End Date Time	Location	
Sensor: S10	Subject: Happy	June 27, 2018, midnight	June 29, 2018, midnight	Neck	[Edit] [Delete]

Figure 3: Deployment object depicting the relationship between a sensor, subject, and date range.

3.2 Data Management

The data management module enables users to upload and automatically parse large amounts of data. It uses parsing information for each sensor type to extract unique identifiers, correctly allocating each file to the appropriate sensor. Time information is extracted from EXIF fields in media files or time data in sensor header files. Users can select multiple files via the file picker UI or upload a zip file containing files for a single sensor. A dropdown menu allows users to select the sensor object for the uploaded files. For the examples in this paper, we have used data from horses [11, 12].

3.3 Sensor Synchronization

We use the device's real-time clock (RTC) for syncing, which timestamps each video and sensor data file. Coarse-grained synchronization is achieved by parsing time information from the data and adding an optional coarse offset. If the RTC clock was not set correctly during data acquisition, a coarse offset can be added on either the sensor page (Figure 2) or the synchronization page (Figure 4). Fine-grained synchronization, detailed at the end of this section, involves annotating multiple offsets for each sensor pair, which are automatically parsed and averaged to achieve sub-framerate precision.

Figure 4 shows the sensor synchronization module that allows the user to select data for synchronization. This page displays the files' parsed start and end times and the manual offset that was applied.

We recommend the following procedure for collecting data for sensor synchronization. This should be performed at least once. When recording over more extended periods, clock drift may occur, and we recommend repeating the procedure at least once per week.

- (1) Start recording video with all cameras simultaneously. Let it run at least a minute before step 3 to ensure overlap with the IMU data.

Select the data that is intended for the synchronization of the sensors

Sensor Data	Begin datetime	End datetime	Manual Offset
☐ Name: DATA-003.CSV Sensor: Sensor: S10	2018-06-28 08:49:07	2018-06-28 08:59:18	0 [Edit]
☐ Name: 20180628_21-59-27.MP4 Sensor: Sensor: Camera A	2018-06-28 19:59:26	2018-06-28 20:09:42	0 [Edit]

Figure 4: Sensor synchronization task generation: After selecting files for synchronization, overlapping segments are automatically fused for offset annotation.

- (2) Enable all IMUs and stack them together. Ensure their IDs are visible to all cameras by tagging or coloring them.
- (3) While keeping the sensors in view, create recognizable patterns by shaking or tapping the devices: (a) Tap once, (b) wait 5 seconds, (c) tap twice, (d) wait 5 seconds, (e) tap three times, (f) place the IMUs on a flat surface and continue recording for another minute.
- (4) Stop recording IMU data.
- (5) Wait 30 seconds, then stop the video.

When the user clicks the "Generate offset annotation task" button, overlapping segments are fused to create an offset annotation task. The user can then navigate to this task and annotate the offset, as shown in Figure 5. Offsets are displayed as purple regions in the time series and can be positive or negative. Multiple distinct moments in the video can be used for annotation, and these annotations are averaged to achieve higher precision.

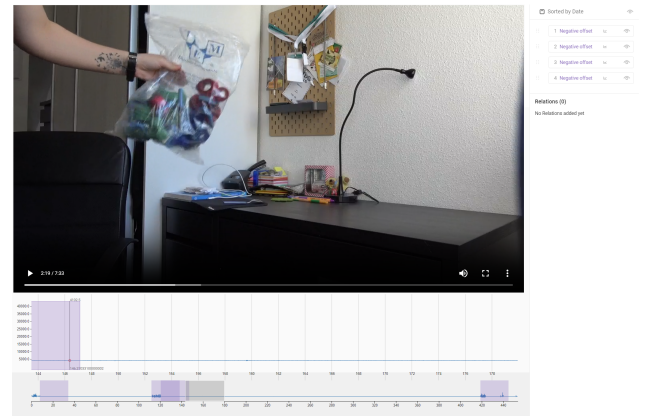


Figure 5: All IMU sensors are placed in a bag and shaken in front of the project's cameras. Various distinct moments captured in the video data are annotated in the IMU data and averaged to enhance offset accuracy.

3.4 Subject Presence Annotation

The subject presence annotation module is used to annotate video segments for subject presence. As shown in Figure 6, videos may contain multiple subjects simultaneously, and subjects may only be present for parts of the videos. M-MOVE-IT automatically populates labels for these annotation tasks using the subject information described in Section 3.1.

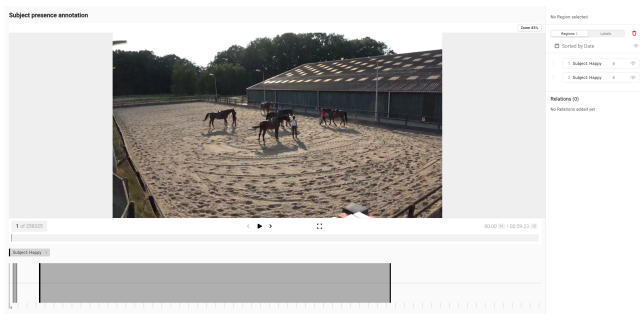


Figure 6: Subject presence annotation interface. This video timeline annotation task depicts when a subject is visually present in a video.

3.5 Activity Task Generation and Annotation

Overlapping video and IMU data segments can be long. Shorter segments are easier to handle on most systems due to internet bandwidth and computer specifications. They also improve precision when using active learning prioritization and are easier to distribute among team members. IMU data files typically contain columns representing sensor data and their axes. The activity task generation interface, shown in Figure 7, allows users to select a subject, set the desired duration for each task, and choose a column from the IMU data to display alongside the video. Our examples use the l^2 -norm derived from the accelerometer axes, denoted as A3D.

Generate activity annotation tasks

Subject:

Subject: Happy

Choose the subject for which tasks should be generated

Segment duration:

300

Choose the duration of each annotation segment in seconds

Column name:

A3D

Generate tasks

Figure 7: Activity task generation interface. The user selects a subject and determines the desired length for the tasks.

The user can navigate to the activity annotation tasks and start annotating activities as a time-series annotation task. The selected column is displayed under the video, and when the grey box in the bottom graph is moved, the video is synchronized to the left edge of the grey box. This allows the user to easily annotate large chunks of the time-series data using the video as ground truth. Due to synchronization, annotations apply to the video and the IMU data. The Label Studio ML back end and API can be used to infer the annotation tasks and add pre-labels that the user can accept or edit. Annotation tasks can be prioritized using the prediction scores of the used algorithms.

3.6 Export data

The export module collects all annotation information in the project and provides a downloadable zip archive. The archive includes annotations and segmented data files, making it straightforward to use

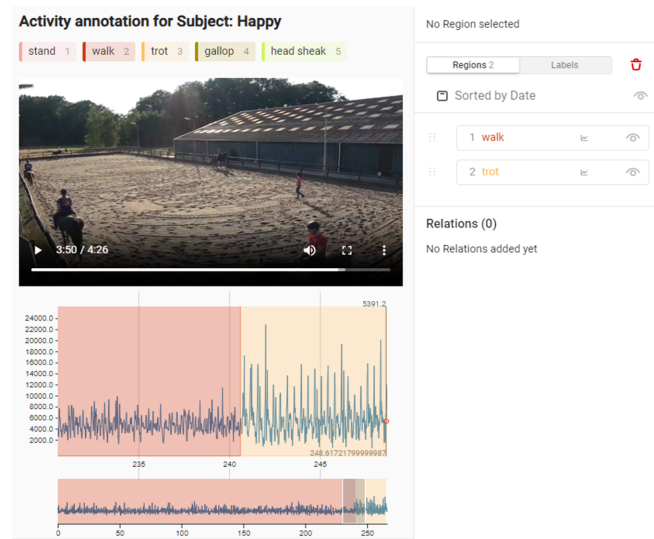


Figure 8: Multimodal activity annotation interface. The synchronized data is displayed together, and the time series can be annotated with activity labels.

the data for subsequent analysis and AI training. The annotations are provided in the JSON format [13] specified by Label Studio.

4 Future Work and Conclusion

M-MOVE-IT is a work in progress, with opportunities for refinement and community-driven expansion. While AI integration for ML-assisted labeling and active learning has been demonstrated locally, it is not yet integrated. However, Label Studio’s documentation makes this accessible to Python users. A module for detecting, classifying, and tracking subjects in videos can automate subject presence annotation. Sensor offset annotation can be automated by visually detecting taps on the IMU sensor in videos. We used a JavaScript workaround recommended for Label Studio versions up to 1.4 for multimodal annotation. The current version, 1.12, includes a new multi-domain template that replaces our workaround, ensuring compatibility with the latest Label Studio version. Additionally, UI and UX improvements are planned. We are excited to demonstrate the framework and welcome community feedback, suggestions, and pull requests.

In conclusion, M-MOVE-IT enhances wearable technology data acquisition, annotation, and AI training by solving video and IMU data synchronization challenges. This open-source framework automates processes like managing sensors, subjects, deployments, and generating activity annotations. It ensures precise synchronization using real-time clocks and offset annotation and automatically parses sensor metadata via a shared repository. The export module integrates annotated data into AI training workflows. M-MOVE-IT supports active learning and human-in-the-loop development, aiding the creation of advanced multimodal AI models and making it valuable for researchers and developers in healthcare, sports, and wildlife monitoring.

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